

A star is born

Clusters of stars are constantly being born from clouds of dust and gas thousands of times the size of our solar system, in a process that can take millions of years.

Born in a cloud

Between existing stars, there are patches of dust and gas. Gradually, these draw in more and more dust and gas to form huge clouds called nebulae. Clumps of matter gather together in these clouds.



Nebula

The process of star formation captured by the *Hubble* telescope.

With enough matter, this process continues. The star gets denser and hotter. Eventually nuclear fusion begins, releasing huge amounts of heat and light: a star is born.

Hot colours

As this matter gets more and more dense, heat builds up to form a young star that fills the surrounding nebulae with light and colour. This spectacular effect (right) was captured by the *Spitzer* space telescope.



Like many space pictures, this image of the Eagle nebula has been artificially coloured so it can be seen clearly.